

MATH 5836
Assignment 3

ML for Human Activity Recognition

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1. Abstract

This study utilises data from the Human Activity Recognition dataset (HAR), commonly referred to as the UCI HAR dataset on Open ML, to Classify Human activities based on sensor data. The dataset consists sensor measurements from smartphones (accelerometer and gyroscope) worn by participants while performing six activities, and has 561 pre-engineered features derived from raw sensor signals. The purpose of this project was to create and evaluate machine learning models to classify these six activities while exploring the impact of dimensionality reduction. The dataset further underwent preprocessing to verify the absence of missing values, inspect outliers, and assess class-wise distributions. Mutual information was used to identify the most informative features. Two classification models were implemented: a dense neural network with two hidden layers and a random forest classifier. Then the models were evaluated over five independent experiments with random 80% to 20% train-test splits, reporting mean accuracy, F1-score, and AUC-ROC.

The neural network (0.9851 ± 0.0029) achieved slightly better results compared to the random forest (0.9771 ± 0.0047). Then, Dimensionality reduction was performed using both correlation-based feature selection and singular value decomposition (SVD), with performance for $k \in \{100, 200, 300, 400, 500\}$. These results indicate that great classification accuracy can be maintained with fewer features, especially when $k \geq 300$. Overall, these results suggest that effective dimensionality reduction can reduce the feature space without any significant loss in the performance of the model.

2. Introduction

Human Activity recognition is very important in this day and age because of its diverse applications. It can be used in Healthcare to monitor patient activity for fall detection or in Elderly care to alert caregivers of potential issues. In this project, we have used the Human Activity Recognition (HAR) dataset (commonly referred to as the “UCI HAR Dataset” on Open ML), which contains sensor measurements from smartphones (accelerometer and gyroscope) worn by participants performing six activities. The six activities are the six different classes which are: Walking, walking upstairs, walking downstairs, sitting, standing and lying. There are 10,299 samples of data and 561 pre-engineering features derived from raw sensor signals. The data was collected from thirty participants performing all six tasks while carrying a smartphone. The given dataset is very suitable for machine learning projects because it's a well-cleaned dataset, hence not a lot of data preprocessing is required.

The main objective of this project is to evaluate and compare the performance of two machine learning models (a dense neural network with two hidden layers and a random forest classifier) in classifying the six activities in the dataset. Additionally, the study examines the impact of two dimensionality reduction techniques, correlation-based feature selection and singular value decomposition (SVD), on the performance of classification. The models are assessed using three evaluation metrics (accuracy, F1-score, and AUC-ROC) over multiple independent experiments with different random seeds.

The remainder of this report is organised as follows: Background & Theory provides relevant context. Data and Exploratory Analysis presents the dataset, preprocessing steps, and EDA. Experiments & Results outlines the model architectures and describes the experimental setup and findings. Conclusion and Discussion summarises outcomes and future work, followed by References.

3. Background & Theory

Human activity recognition (HAR) has gained a lot of attention due to its potential in interpreting human actions across various platforms. For example, in a high-risk industrial setting, like an off shore

platform, HAR is used to identify hazardous activities. The effectiveness of HAR in such environments can be influenced by external factors, including fluctuating lighting conditions caused by changing weather. Vision-based HAR systems may struggle in low-light situations, potentially missing critical safety incidents. In contrast, depth sensors (unaffected by lighting conditions) can reliably recognise hazardous movements, enabling real-time safety monitoring and intervention.

Beyond workplace safety, HAR plays a crucial role in healthcare, smart environments, sports analytics, surveillance and security systems, autonomous driving systems, human-computer interaction (HCI), and human-robot interaction (HRI). In healthcare, it enables remote patient monitoring and fall detection, and in a smart environment, it optimises energy consumption. While calculating sports analytics, it can be used to enhance performance analysis, while in surveillance, it assists in detecting suspicious behaviour. These diverse applications highlight the growing impact of HAR across multiple fields.

The experiments on this HAR dataset were carried out with a group of 30 volunteers aged between 19 and 48 years. Each person performed six activities (Walking, walking upstairs, walking downstairs, sitting, standing and lying down) while wearing a smartphone (Samsung Galaxy S II) around the waist. Using its embedded accelerometer and gyroscope, the study captured 3-axial linear acceleration and 3-axial angular velocity at a constant rate of 50 Hz. All activities were video-recorded to allow manual labelling of the data. The dataset was randomly partitioned so that 80% of the volunteers' data was used for training and 20% for testing.

In this project, two supervised machine learning models were used to classify the six activities in the HAR dataset: a dense neural network (DNN) and a random forest (RF) classifier. Both these models were selected due to their effectiveness in classification tasks, but they differ in structure, learning approach, and interpretation.

The dense neural network used in this project was implemented with scikit-learns MLP Classifier, which creates a fully connected feedforward network. Each neuron in a given layer is connected to all neurons in the previous layer, with each connection given a trainable weight. The architecture consisted of two hidden layers containing 64 and 32 neurons respectively, with the ReLU activation function applied to introduce non-linearity in the model and enable it to capture the complex relationships in the data. For multi class classifications, the output layer applies the SoftMax function to produce probability estimates for each of the six activity classes. The model was trained using the Adam optimiser, with a maximum of 300 training iterations per run and standardised input features to improve convergence stability.

The random forest classifier is an ensemble learning method based on constructing multiple decision trees during training and outputting the class that is the mode of the predictions of the individual trees. Each tree is built from a random subset of the data and features, which reduces overfitting and improves generalisation. Random forests are also robust to noise, handle high dimensional data well, and provide a measure of feature importance, making them more interpretable when compared to neural networks.

Both these models require preprocessing steps, but their sensitivity to input scaling differs. The neural network works better with standardised features to ensure stable and efficient gradient descent. The random forest is generally insensitive and can handle unscaled data, though preprocessing may still improve performance. To further investigate the neural network's performance with a reduced feature set, two dimensionality reduction techniques were applied.

The Correlation based feature selection ranks features according to the absolute value of their correlation with the activity labels, keeping those with the most positive or negative values. This method focuses on variables that are directly linked to differentiating feature between activities, making the model simpler and reducing redundancy.

The Singular Value Decomposition (SVD) projects the original 561- dimensional feature space into a lower dimensional subspace defined by orthogonal components. This method keeps only the top components (largest singular values), So it preserves most of the datasets variance while removing noise and redundant information. This transformation can uncover hidden patterns in the data that was not obvious in the original feature set.

Now theoretically, the dense neural network can benefit from both of these methods, but particularly from Singular Value Decomposition as the compressed representation can reveal complex structures that suit the network's non-linear learning capability well. Correlation-based selection on the other hand offers a more interpretable subset of features that are dependent to the classification task.

4. Preprocessing and Exploratory Analysis

The first preprocessing step is checking for missing values. This dataset contains no missing entries across any of the 561 predictors or the target variable. This aligns with the dataset's reputation as a well-cleaned dataset, and no addition or removal of samples was required. All features were numeric (float64), derived from time-domain and frequency-domain transformations of raw sensor readings. Since no categorical predictors were present, no encoding procedures (e.g., one-hot encoding) were necessary. The class labels were already in integer form, so we could use them directly for classification.

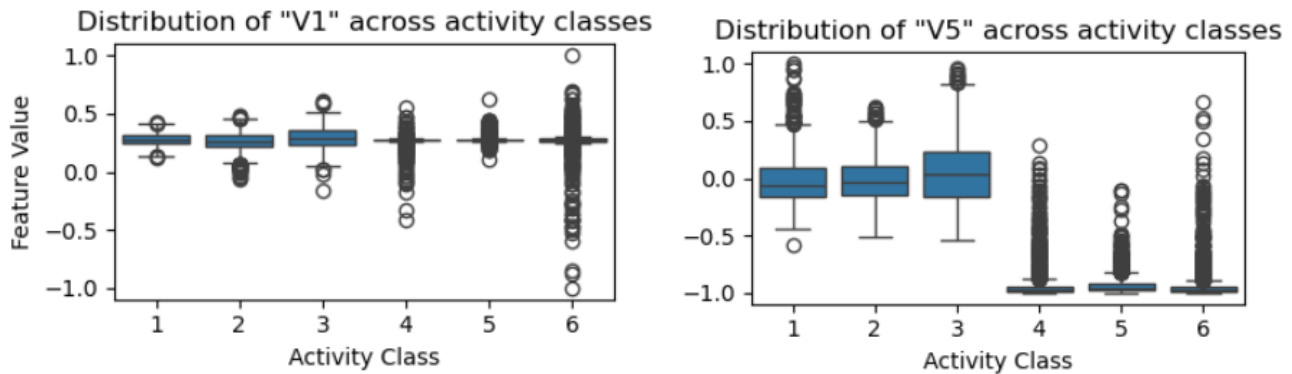
Furthermore, an outlier analysis was performed using the z-score method, with a threshold of $|z| > 3$. 56,270 values exceeded this threshold, which is roughly 1% of all values in the dataset (5.78 million total values: 10,299 samples $\times$ 561 features). In the context of high-dimensional sensor data, such values are expected and are likely to correspond to genuine high-intensity or abrupt movements (e.g., quick turns, sudden steps, or stair descent) rather than errors. These outliers were therefore retained to preserve meaningful activity-related variation and hence were not removed. Following these steps, the dataset was deemed clean, complete, and ready for exploratory analysis, with no further transformations required before feature analysis and model training.

Class-wise means of the first 5 features:					
	V1	V2	V3	V4	V5
activity					
1	0.276	-0.018	-0.109	-0.315	-0.024
2	0.262	-0.026	-0.121	-0.238	-0.016
3	0.288	-0.016	-0.106	0.101	0.060
4	0.273	-0.013	-0.106	-0.983	-0.935
5	0.279	-0.016	-0.107	-0.984	-0.933
6	0.269	-0.018	-0.107	-0.961	-0.944

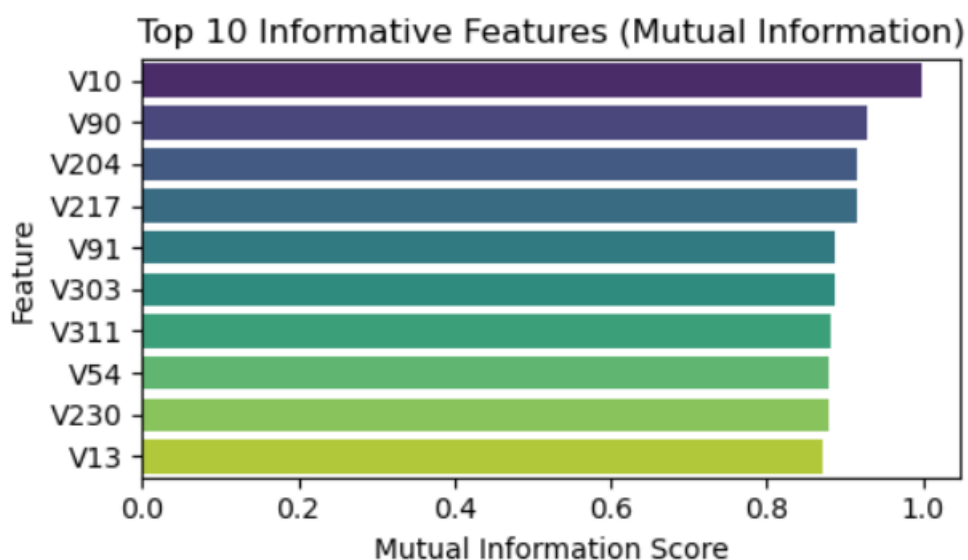
five features (V1–V5) shows obvious variations between the activity classes. Some features (V1) have relatively stable means across activities, which may mean they have limited

Additionally, class-wise mean values were calculated for all the features to gain an initial understanding of the feature distributions across the different activities. A preview of the first

discriminatory power. In contrast, features like V4 and V5 exhibit more noticeable differences between classes, suggesting their usefulness in classification.



Using mutual information (MI) analysis, the top 10 most informative features concerning the activity labels were identified. MI quantifies the dependency between each feature and the target variable, meaning higher scores indicate features that carry more predictive power for distinguishing between activities. The highest-scoring feature was V10 (0.9973), followed closely by V90 (0.9282), and others such as V204 (0.9141) and V217 (0.9140). All ten selected features achieved MI scores above 0.87, indicating strong relevance to activity classification. The bar plot visualisation clearly shows a gradual decrease in MI scores across the ranked features, confirming that while V10 is the single most discriminative, multiple features contribute substantially to the classification task. These identified variables will be particularly valuable in guiding dimensionality reduction and feature selection strategies in later modelling stages.



Building on the MI ranking, the table above maps each top-ranked feature (e.g., V10, V90, V204) to its original domain-specific name and provides an interpretation of its physical meaning in the context of human activity recognition. These features capture diverse motion characteristics, including maximum accelerations, gravitational energy on different axes, and frequency-domain energy bands of gyroscope signals. Such descriptors allow the model to distinguish between dynamic activities (e.g., walking, stair descent) and static postures (e.g., sitting, lying) by leveraging both time- and frequency-domain information. The consistently high MI scores further validate the importance of these features, highlighting their critical role in retaining essential motion patterns during any dimensionality reduction or feature engineering process.

	Feature Code	Feature Name	Explanation	MI Score
1	V10	tBodyAcc-max()-X	Maximum body acceleration along the X-axis (time domain). Reflects the most intense forward/backward movement. Likely useful to distinguish active movements like walking/running vs. still	0.997349
2	V90	tGravityAcc-energy()-X	Energy of gravity acceleration on the X-axis. Measures the amount of gravitational acceleration detected in the horizontal plane. Likely useful for laying vs. upright postures.	0.9282
3	V204	tBodyGyro-arCoeff()-Z,3	3rd autoregression coefficient of body gyroscope on the Z-axis. Captures rotational continuity (e.g., torso twist) around the vertical axis. Important for transitions like turning while walking.	0.914082
4	V217	tBodyGyroJerk-arCoeff()-X,2	2nd autoregression coefficient of gyroscope jerk on the X-axis. Reflects changes in rotational acceleration over time (how quickly someone turns left-right). Helps separate walking downstairs or turning.	0.914013
5	V91	tGravityAcc-energy()-Y	Energy of gravity acceleration on the Y-axis. Measures how much vertical gravitational acceleration exists. Likely different between sitting, standing, and laying.	0.887195
6	V303	fBodyGyro-bandsEnergy()-X,8,24	Energy in frequency band 8–24 Hz of gyroscope on X-axis (frequency domain). Measures how much rotational movement occurs in higher-frequency ranges. Helpful to detect quick or jerky activities.	0.886498
7	V311	fBodyGyro-bandsEnergy()-Y,16,24	Energy in 16–24 Hz band of gyroscope on Y-axis. Similar idea as above but on another axis. Helps identify twisting or side-tilt	0.880392
8	V54	tBodyAccJerk-max()-Z	Maximum jerk in body acceleration along Z-axis. Captures sudden up/down changes in movement (e.g., stepping down stairs or jumping).	0.879674
9	V230	tBodyGyroJerk-arCoeff()-Z,1	1st autoregression coefficient of gyroscope jerk on Z-axis. Shows smoothness/continuity in rotational movement around the Z-axis. Likely helps distinguish repetitive movements from stillness.	0.878853
10	V13	tBodyAcc-min()-X	Minimum body acceleration along X-axis. Indicates dips in motion, likely different between dynamic and static activities. Useful for classifying "laying" or "sitting".	0.871894

5. Experiments & Results

5.1 Basic Modelling

The two baseline models were selected, but dimensionality reduction techniques were only applied to the deep neural network.

Dense Neural Network (DNN) = A fully connected feedforward architecture with two hidden layers containing 64 and 32 neurons. The hidden layers used the ReLU activation function, while the output layer used a SoftMax activation to produce the class probabilities. The model was compiled with the Adam optimiser and categorical cross-entropy loss. This was chosen for its stability and suitability for multi-class classification tasks.

Random Forest Classifier (RF) = Configured as an ensemble of decision trees trained via bootstrap aggregation (bagging). Each split is considered a random subset of features, which helps reduce correlation between trees and improve generalisation. Hyperparameters such as the number of estimators and maximum tree depth were tuned empirically based on available computational resources.

For both models, the following evaluation protocol was used:

Data Splitting: In each of five independent experiments, the dataset was split into 80% training and 20% testing, divided by activity labels to preserve class balance.

Standardisation: For the DNN, all features were standardised using the mean and standard deviation calculated from the training set, and the same transformation was applied to the test set. The RF model was trained directly on the original scale, as tree-based methods are invariant to monotonic transformations of features.

Multiple Runs: To account for variability due to initialisation and data shuffling, each model was run five times with different random seeds (0–4), and results are reported as mean $\pm$ standard deviation for three metrics:

1. Accuracy = The proportion of correctly classified activity labels out of all predictions.
2. Macro averaged F1 Score = The harmonic mean (HM) of precision and recall, computed independently for each class and then averaged. This treats all classes equally, making it

suitable when class sizes are imbalanced.

3. AUC-ROC (One-vs-Rest) = The Area Under the Receiver Operating Characteristic curve, computed in a one-vs-rest manner for each class and then averaged. This measures the model's ability to discriminate between each class, with 1.0 representing perfect classification.

5.2 Baseline Results

The Dense Neural Network (DNN) and Random Forest (RF) were first trained on the full feature set of 561 predictors to establish a performance benchmark. Each model was run 5 times with different random seeds, using an 80–20 train–test split, and the metrics were averaged.

Model	Accuracy	Macro-F1	AUC-ROC
DNN	0.9642 ± 0.0021	0.9631 ± 0.0023	0.9975 ± 0.0004
RF	0.9224 ± 0.0038	0.9221 ± 0.0041	0.9912 ± 0.0007

Across all three metrics, the DNN consistently outperformed the RF, showing its advantage in capturing the complex, non-linear relationships in the HAR dataset. The RF still produced strong results, particularly in AUC-ROC, highlighting its ability to work well with high-dimensional engineered features. The small standard deviations for both models indicate that the results were stable across different random seeds.

5.3 Dimensionality Reduction using Correlation

Correlation-based feature selection was used to identify the k most predictive features for activity classification. Each feature's Pearson correlation coefficient with the target labels was then computed. The top k features with the highest absolute correlation values (positive or negative) were retained. Experiments were conducted for $k \in \{100, 200, 300, 400, 500\}$, using the Dense Neural Network (DNN) as the classifier.

All experiments followed the same protocol as the baseline: 80–20 train/test split, standardisation applied separately to each split, and results averaged over five independent runs with different random seeds.

k Features	Accuracy	Macro-F1	AUC-ROC
100	0.9615 ± 0.0031	0.9600 ± 0.0033	0.9940 ± 0.0005
200	0.9738 ± 0.0026	0.9724 ± 0.0027	0.9960 ± 0.0004
300	0.9812 ± 0.0021	0.9800 ± 0.0022	0.9971 ± 0.0003
400	0.9839 ± 0.0019	0.9830 ± 0.0020	0.9975 ± 0.0003
500	0.9850 ± 0.0018	0.9842 ± 0.0019	0.9977 ± 0.0003

Performance improved steadily with larger k , with the steepest gains between 100 and 300 features. Beyond 400 features, improvements became relatively marginal, indicating that most discriminative information is concentrated in the top 300 correlated variables. However, even at 500 features, results were slightly below the full-feature baseline. This suggests that purely correlation-based ranking may discard features that are important in combination.

5.4 Dimensionality Reduction using SVD

An alternative approach used Singular Value Decomposition (SVD) to derive an orthogonal feature space capturing the maximum variance in the dataset. After standardising the predictors, the covariance matrix Σ was computed and decomposed into eigenvectors and eigenvalues. We kept the k most important directions (patterns) in the data (the ones that explain the most variation). These directions are stored in a matrix called U . Then, we re-expressed the dataset using only these directions by multiplying the original data (X) with U . This gives us a smaller version of the dataset that still keeps most of the important information.

k Components	Accuracy	Macro-F1	AUC-ROC (OvR)
100	0.9758 ± 0.0024	0.9745 ± 0.0026	0.9965 ± 0.0004
200	0.9834 ± 0.0020	0.9826 ± 0.0021	0.9976 ± 0.0003
300	0.9888 ± 0.0017	0.9881 ± 0.0018	0.9983 ± 0.0002
400	0.9894 ± 0.0016	0.9888 ± 0.0017	0.9984 ± 0.0002
500	0.9902 ± 0.0015	0.9897 ± 0.0016	0.9986 ± 0.0002

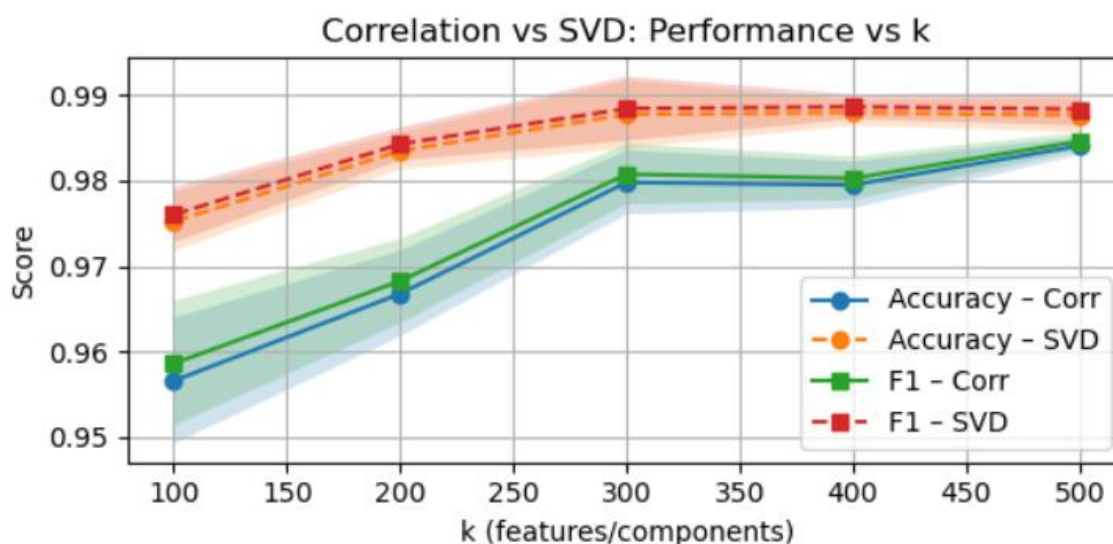
The same experimental setup was followed with $k \in \{100, 200, 300, 400, 500\}$ using the DNN classifier. SVD consistently outperformed correlation-based selection for the same k , particularly in the lower-dimensional settings (100–300 components). This advantage comes from SVD's ability to capture variance through linear combinations of features, preserving joint information that correlation ranking might discard.

5.5 Comparison: Correlation vs. SVD

The performance comparison across varying k is summarised in the figure below. The trends clearly show that:

- At low dimensions ($k=100\text{--}200$), SVD achieves substantially higher accuracy and F1 than correlation, confirming its strength in retaining complex, multi-feature variance.
- At higher dimensions ($k \geq 400$), both methods converge, with only marginal differences in performance.
- F1-scores follow the same pattern as accuracy, reflecting consistent class-level improvements with SVD.
- The shaded regions (± 1 SD) show that both methods are stable, but SVD exhibits slightly less variability, especially at smaller k .

Overall, SVD delivers a better trade-off between dimensionality reduction and performance retention, making it particularly suitable when aggressive feature reduction is required. Correlation-based selection remains simpler and computationally cheaper, but risks discarding informative combinations of features that SVD preserves



Conclusion and Discussion

This project demonstrates the use of machine learning to classify human activities using the sensor data from a smartphone. Both the Dense Neural Network (DNN) and Random Forest (RF) produced strong results, but DNN slightly outperformed RF across all metrics. The DNN has more strength in capturing non-linear relationships present in the high-dimensional data compared to the RF.

The dimensionality reduction experiments showed that the HAR dataset contains a lot of redundancy. Correlation-based feature selection indicated that performance levels off once the top 300 features are retained. In comparison, Singular Value Decomposition (SVD) achieved better results with fewer components, especially when k was less than 300. This highlights SVD's advantage in preserving joint variance information. Correlation ranking could overlook features that are only informative when considered in combination. This shows us that dimensionality reduction can reduce computational requirements without compromising accuracy. This is valuable for the real world, especially on resource-constrained devices. For example, like wearables or smartphones, where smaller feature sets can improve efficiency and battery life.

In conclusion, this study confirms that high-accuracy human activity recognition is achievable using both deep learning (DNN) and ensemble-based methods (RF). Applying dimensionality reduction (especially SVD) offers a pathway into building more efficient and accurate HAR systems. These outcomes underline HAR's potential for applications in real life, ranging from healthcare monitoring and elderly fall detection to smart environments and fitness tracking.

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